MediCore
HEALTH SYSTEMS PVT. LTD.
Care, Connected.

MEDICORE HEALTH SYSTEMS PVT. LTD.

Registered Office: Plot 42, HITEC City, Hyderabad, Telangana – 500 081, India
CIN: U85100TG2012PTC078342 | GST: 36AABCM1234P1ZV

PROBLEM DIAGNOSTIC DOCUMENTATION

Oracle ERP Data Architecture Assessment
& Data Warehouse Migration Roadmap

Version 1.0 ■ June 2025 ■ Data Architecture & Analytics Division

COMPANY	DOCUMENT TYPE	DATE	CLASSIFICATION
MediCore Health Systems Pvt. Ltd.	Problem Diagnostic Documentation	June 2025	**CONFIDENTIAL**

PREPARED BY	REVIEWED BY	APPROVED BY
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DOCUMENT PURPOSE

This document assesses the current Oracle ERP data architecture at MediCore Health Systems, quantifies the operational and financial impact of the existing flat-table design, and defines a phased roadmap to a normalised dimensional warehouse with a modern analytics layer.

1. Company Overview

MediCore Health Systems Pvt. Ltd. is a multi-specialty hospital network headquartered in Hyderabad, Telangana. Founded in 2012, the company operates seven hospital facilities across Andhra Pradesh and Telangana, with a combined capacity of over 1,400 beds and a network of more than 850 registered medical practitioners.

Legal Name	MediCore Health Systems Pvt. Ltd.
Incorporation	2012 ■ Registered under Companies Act 2013
CIN	U85100TG2012PTC078342
Headquarters	Plot 42, HITEC City, Hyderabad, Telangana – 500 081
Facilities	7 hospitals — Hyderabad, Vijayawada, Warangal, Karimnagar, Nellore
Total Beds	1,400+ across all facilities
Medical Staff	850+ doctors; 2,200+ nurses and paramedical staff
Annual Patients	4.2 lakh outpatients + 68,000 inpatients (FY2024–25)
Revenue (FY2024–25)	**Rs. 412 crore** (consolidated)
Core Business	Cardiology, Neurology, Oncology, Orthopedics, General Medicine, Pediatrics, Nephrology, Endocrinology
ERP System	Oracle Health Cloud + Oracle ERP Finance (since 2018)
Analytics Platform	Legacy: Oracle BI Publisher (being replaced)
Regulatory	NABH Accredited ■ ISO 9001:2015 ■ HIPAA-aligned ■ DPDP Act 2023 compliant

1.1 Organisational Structure

MediCore operates through three business verticals: Clinical Operations (patient care delivery), Administrative Services (billing, HR, procurement), and Health Analytics (outcomes research and performance reporting). The Data Architecture & Analytics Division falls under Administrative Services and reports to the CTO.

1.2 Technology Landscape

- **Oracle Health Cloud** — clinical records, appointment scheduling, lab integration
- **Oracle ERP Finance** — billing, accounts payable/receivable, GST compliance
- **Oracle SCM** — pharmacy procurement and medical supply chain
- **Oracle HCM** — human resources, payroll, staff scheduling
- **Legacy Oracle BI Publisher** — standard operational reports (being sunset)
- **Microsoft 365** — collaboration, email, SharePoint document management

2. Problem Statement

MediCore Health Systems has identified five critical operational and data management failures that are directly impacting patient safety, financial performance, and regulatory compliance. These problems stem from fragmented data architecture, over-reliance on a single flat Oracle ERP table, and the absence of a unified analytics platform.

EXECUTIVE SUMMARY OF THE CORE PROBLEM

All clinical, administrative, billing, and insurance data from 7 hospital facilities is being written into a single flat Oracle ERP table — **hospital_raw** — containing 49 columns and over 5,000 records per month. This denormalised structure creates cascading data quality failures, prevents real-time clinical decision support, and has resulted in **Rs. 18 lakh per month in billing disputes** and a **28% insurance claim rejection rate**.

2.1 Problem 1 — Flat Table Architecture (CRITICAL)

Problem Description

The entire hospital data ecosystem — patient demographics, doctor records, department information, appointment details, lab results, prescriptions, billing, and insurance — is stored in a single denormalised Oracle ERP table. This violates First, Second, and Third Normal Form (1NF, 2NF, 3NF) and creates structural data integrity failures.

Root Cause Analysis

- Oracle ERP implementation team chose a flat table design in 2018 for simplicity of initial deployment, without considering long-term scalability or data quality requirements.
- No data modelling standards were enforced during ERP configuration — all modules write directly to hospital_raw without staging or validation layers.
- Absence of a data governance framework meant the flat table design went unchallenged for 7 years.

Measurable Impact

METRIC	CURRENT STATE	BUSINESS IMPACT
Data redundancy	Patient name repeated in every appointment row	Storage waste; update anomalies
Update anomaly	Doctor fee change requires updating 1,000s of rows	Data inconsistency across records
Insert anomaly	Cannot register patient without an appointment	Operational bottleneck
Delete anomaly	Deleting last appointment deletes all patient data	**Irreversible data loss risk**

2.2 Problem 2 — Billing Discrepancies (CRITICAL)

Problem Description

Invoices are auto-generated from the flat table without validating that **total_amount = consultation_fee + lab_charge + medicine_charge** – **discount + GST**. As of May 2025, billing discrepancies are costing MediCore **Rs. 18 lakh per month** in patient disputes and write-offs.

Root Cause Analysis

- No automated assertion checks in the Oracle ERP billing module — amounts are written to hospital_raw without cross-validation.
- GST computation errors in 14% of invoices due to incorrect tax codes applied by the ERP for different service categories.
- Discount application is manual and not enforced by business rules — discounts exceeding the subtotal have been processed.
- No audit trail for post-facto invoice modifications — manual corrections are not logged in the ERP.

ISSUE	FREQUENCY	AVG VALUE (INR)	MONTHLY LOSS (INR)
GST mismatch	14% of invoices	1,250	**6,30,000**
Discount > subtotal	3% of invoices	3,800	**4,20,000**
Incorrect service code	8% of invoices	2,100	**7,50,000**
TOTAL MONTHLY LOSS			**18,00,000**

2.3 Problem 3 — Insurance Claim Failures (CRITICAL)

Problem Description

28% of all insurance claims submitted to providers (Star Health, HDFC ERGO, ICICI Lombard, Bajaj Allianz) are being rejected. The primary cause is incorrect or missing data in the claim submission package, which is auto-generated from the flat hospital_raw table without data validation. **Annual financial impact: Rs. 45 lakh.**

Rejection Reasons Breakdown

REJECTION REASON	% OF REJECTIONS	ROOT CAUSE IN ERP
Missing or incorrect ICD-10 diagnosis code	**34%**	diagnosis field not mapped to ICD-10
Pre-existing condition not flagged	**22%**	allergies/history not in submission
Policy number mismatch or expired	**19%**	policy_number not validated at admission
Invoice amount vs claim amount discrepancy	**15%**	billing errors from Problem 2
Missing lab results in support documents	**10%**	lab system not linked to ERP claim

2.4 Problem 4 — No Real-Time Lab Integration (HIGH)

Problem Description

Laboratory test results are recorded in hospital_raw with an average delay of **24 to 48 hours** after test completion. Results arrive on paper or via email and are manually entered into the ERP by administrative staff. There is no digital integration between the lab equipment, the Oracle ERP system, and the treating doctor's clinical dashboard.

- Doctors are making clinical decisions without the latest test data in **31% of inpatient cases**, as reported by the clinical audit committee in Q1 2025.
- Manual data entry introduces transcription errors — a 2024 internal audit found a **6.2% error rate** in lab result entries.
- Critical abnormal results (is_abnormal = TRUE) have no automated alert mechanism — doctors must actively check the system.
- Turnaround time for high-priority tests (troponin, D-dimer, CBC post-chemo) should be under 2 hours but averages **28 hours** due to manual workflow.

2.5 Problem 5 — Fragmented Analytics & Reporting (HIGH)

Problem Description

MediCore has no unified analytics platform. Each department generates reports independently from the flat Oracle ERP table using Oracle BI Publisher (scheduled for decommissioning in Q4 2025). Cross-functional reports — for example clinical outcomes against billing and insurance performance — are impossible to generate in real time.

- Revenue reports take **3 to 5 business days** to compile because data from 7 facilities must be manually consolidated.
- No Power BI or equivalent dashboards — hospital leadership makes strategic decisions based on reports that are weeks old.
- Data analysts spend **70% of their time** on data extraction and cleaning rather than insight generation.
- No single source of truth for patient count, revenue, or clinical KPIs across the group.

3. As-Is Data Architecture Assessment

This section documents MediCore's current data architecture — the Oracle ERP flat table design, its structural deficiencies, and the downstream systems affected. All 49 columns in hospital_raw are assessed for normalisation violations, data quality issues, and ETL readiness.

3.1 Current Architecture Overview

Source System	Oracle Health Cloud + Oracle ERP Finance (Version 23.3)
Database	Oracle Database 19c — on-premise, HITEC City Data Centre
Primary Table	hospital_raw — single denormalised table, all domains
Records (2020–25)	5,000+ rows per month; 3,60,000+ total across 5 years
Column Count	49 columns spanning patient, doctor, department, appointment, lab, medicine, billing, insurance
Current Storage	~4.2 TB raw; 1.8 TB post-compression
Backup Policy	Daily RMAN backup to on-premise tape; 7-day retention **(insufficient)**
Reporting Tool	Oracle BI Publisher — being decommissioned Q4 2025
Analytics	None — ad hoc SQL queries only

3.2 Normalisation Violations in hospital_raw

First Normal Form (1NF) Violations

- **Repeating groups:** multiple medicine names crammed into a single medicine_name field, separated by commas in early records.
- **Non-atomic values:** the allergies field contains multi-value strings ('Penicillin, Sulfa drugs') that should be a separate child table.
- **No primary key** enforced at the Oracle ERP table level — record_id is application-assigned but not a database constraint.

Second Normal Form (2NF) Violations

- doctor_name, specialization, consultation_fee and available_days are functionally dependent only on doctor_id, not on the full composite key (appointment_id + patient_id + doctor_id).
- department_name, floor_number and head_name depend only on department_id, yet they repeat in every appointment row for that department.
- insurance_provider, policy_number and coverage_limit depend only on patient_id, not on the appointment.

Third Normal Form (3NF) Violations

- department_head (head_name) is transitively dependent: appointment → department_id → department_name → head_name.
- Consultation fee is stored in the appointment row instead of the doctor entity — changing a doctor's fee requires updating all past appointment records.
- Fiscal year and fiscal quarter are transitively derived from appointment_date via a non-key relationship.

3.3 Data Quality Assessment by Domain

DOMAIN	KEY ISSUES	NULL RATE	ERROR RATE	RISK
Patient	Duplicate patient_ids across facilities; allergies NULL in 12% of records	12% allergies 4% email	6.2% transcription	**CRITICAL**
Doctor	Consultation fee inconsistency — same doctor, different fees in different rows	0%	8.3% fee variance	**HIGH**
Appointment	14 records with appointment_date beyond 6 months in the future; status inconsistencies	0.3%	2.1%	**MEDIUM**
Lab	6.2% of lab results carry a reference_range copied from the wrong test; is_abnormal missing in 9% of completed tests	9% is_abnormal	6.2% result errors	**CRITICAL**
Billing	total_amount ≠ subtotal – discount + gst in 14% of records; negative discount in 3 records	0%	14% total mismatch	**CRITICAL**
Insurance	'None' not consistently used for self-pay; 23 records with policy_number but provider = None	38% self-pay	4.6% inconsistency	**HIGH**

VERIFICATION NOTE

The figures in section 3.3 are as originally assessed. An independent profiling run against the 5,000-row extract has since produced materially different results for several of these metrics — notably the billing mismatch rate and the allergies null rate. Section 3.3 should be reconciled against the source profiling report before this document is circulated externally.

4. To-Be Architecture — Star Schema Target State

The proposed solution replaces the single flat Oracle ERP table with a fully normalised star schema. Data flows from Oracle ERP through a structured Bronze → Silver → Gold medallion pipeline, with dashboards serving clinical, billing, and executive teams.

4.1 Star Schema Design

TABLE	TYPE	GRAIN	KEY COLUMNS
fact_appointment	FACT	1 row per appointment	appointment_sk PK ■ patient_sk FK ■ doctor_sk FK ■ department_sk FK ■ date_sk FK ■ total_amount ■ payment_status
dim_patient	DIMENSION	1 row per patient	patient_sk PK ■ patient_id NK ■ full_name ■ blood_group ■ allergies ■ insurance_sk FK
dim_doctor	DIMENSION	1 row per doctor	doctor_sk PK ■ doctor_id NK ■ specialization ■ consultation_fee ■ department_sk FK
dim_department	DIMENSION	1 row per department	department_sk PK ■ department_id NK ■ dept_name ■ floor_number ■ head_name
dim_date	DIMENSION	1 row per calendar date	date_sk PK ■ full_date ■ day ■ month ■ quarter ■ year ■ is_weekend ■ fiscal_year
dim_lab_test	SUB-DIM	1 row per lab order	lab_sk PK ■ appointment_sk FK ■ test_name ■ result ■ reference_range ■ is_abnormal
dim_prescription	SUB-DIM	1 row per medicine line	prescription_sk PK ■ appointment_sk FK ■ diagnosis ■ medicine_name ■ dosage ■ frequency
dim_insurance	OUTRIGGER	1 row per insurance policy	insurance_sk PK ■ patient_sk FK ■ provider_name ■ policy_number ■ coverage_limit ■ valid_until

4.2 Medallion Architecture Layers

BRONZE LAYER

Raw ingested data from Oracle ERP. An exact copy of hospital_raw, loaded without transformation. Retained permanently for audit, replay, and regulatory evidence. Table: **bronze.hospital_raw**.

SILVER LAYER

Cleaned, validated, type-cast and normalised data. Null handling (allergies → 'None', charges → 0), billing total assertions, is_abnormal Boolean casting, duplicate removal, and surrogate key generation. Dimension tables are created from this layer.

GOLD LAYER

Analytics-ready fact and aggregate tables, pre-joined and pre-aggregated for the reporting layer. Tables: **gold.fact_appointment**, gold.dept_revenue_monthly, gold.abnormal_labs_daily, gold.doctor_performance, gold.claims_aging.

4.3 Ingestion Strategy from Oracle ERP

METHOD	USE CASE	LATENCY	SETUP REQUIRED
CSV Upload (Initial)	One-time full historical load (2020–2025)	One-time	Nil — direct file load
Data Factory Pipeline	Daily incremental sync from Oracle	15–60 minutes	SHIR agent on hospital server
Database Mirroring	Near-real-time replication	~30 seconds	Intermediate SQL database
EventStream (Kafka)	Live lab results streaming	< 5 seconds	Event Hub + Kafka connector

5. Implementation Roadmap

The migration from the Oracle ERP flat table to a dimensional warehouse is planned as a six-phase programme over 16 weeks. Each phase has defined deliverables, owners, and success criteria.

PH	PHASE NAME	KEY DELIVERABLES	DURATION	OWNER
1	**Data Assessment & Design**	Column mapping file, ER diagram, normalisation plan, stakeholder sign-off	2 weeks	Data Architect
2	**Oracle Export & Bronze Load**	Oracle Data Pump export, CSV files for 2020–25, lakehouse provisioned, bronze.hospital_raw loaded	2 weeks	DBA + Data Eng.
3	**Silver Layer — Cleanse & Model**	Notebooks for all 8 dimension tables, data quality assertions, surrogate key generation, 3NF validation	3 weeks	Data Engineering
4	**Gold Layer — Fact & Aggregates**	fact_appointment built with all SK lookups, 4 gold aggregate tables, pipeline scheduled	3 weeks	Data Engineering
5	**Dashboards**	Revenue dashboard, clinical KPI dashboard, claims ageing dashboard	3 weeks	Analytics Team
6	**UAT, Go-Live & Monitoring**	User acceptance testing, parallel run with BI Publisher, cutover, alerting, DPDP compliance sign-off	3 weeks	All teams

5.1 Success Metrics (Post Go-Live KPIs)

KPI	CURRENT BASELINE	TARGET (6 MONTHS)	MEASUREMENT
Billing dispute rate	Rs. 18L / month	**< Rs. 2L / month**	Finance team weekly
Insurance claim rejection rate	28%	**< 5%**	Claims portal report
Lab result delivery (digital)	24–48 hours	**< 2 hours**	Lab system timestamp
Revenue report generation time	3–5 days	**< 30 minutes**	Dashboard refresh log
Data analyst productivity	30% insight time	**80% insight time**	Time tracking survey
Data quality score	61% (assessed)	**> 95%**	Automated DQ notebook
Query response time	N/A (manual)	**< 2 seconds**	Performance log

6. Risk Register

RISK	DESCRIPTION	LIKELIHOOD	IMPACT	MITIGATION
Data loss during migration	Records lost during Oracle export or ingestion	Low	Critical	Checksums; bronze retained as backup
Oracle ERP downtime	ERP unavailable during the export window	Medium	High	Schedule export in maintenance window
Data quality rejection	High rejection rates in silver validation	High	Medium	Quarantine table; manual review workflow
User adoption resistance	Clinical staff resist the new dashboard interface	Medium	Medium	Training programme; champion network
DPDP Act compliance	Patient PII in cloud may require a consent audit	Medium	High	PII masking in silver; DPA audit before go-live
Capacity cost overrun	Compute capacity cost may exceed budget approval	Low	Medium	Start with a smaller pilot; scale after ROI proven

7. Document Control & Approvals

VERSION	DATE	AUTHOR	CHANGE DESCRIPTION
v0.1	15 May 2025	Aravind Seshagiri	Initial draft — problem statement
v0.8	28 May 2025	Aravind Seshagiri	Added as-is assessment and risk register
v1.0	5 June 2025	Preethi Ramachandran	Final review and CTO approval
v1.1	August 2026	Saratchendra Nitturi	Reissued as a formatted document; verification note added to section 3.3

PREPARED BY	REVIEWED BY	APPROVED BY
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